

Rotation and Torque

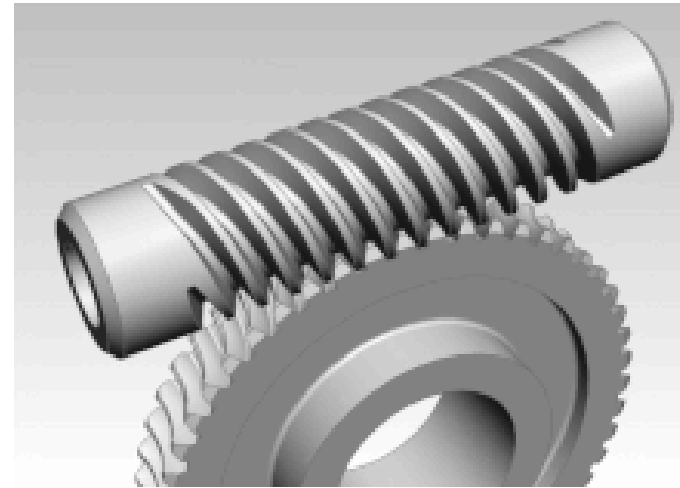
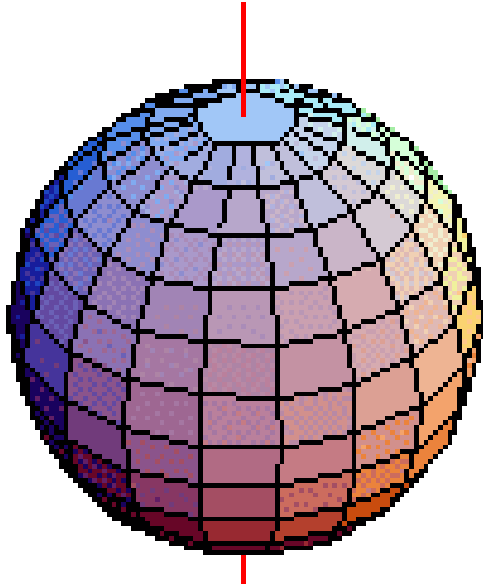
SS ILP Physics (Intermediate), Week 5 Class 4



Outline

- Fixed-axis rotations
- Rotational kinematics
- Moment of inertia
- Torque

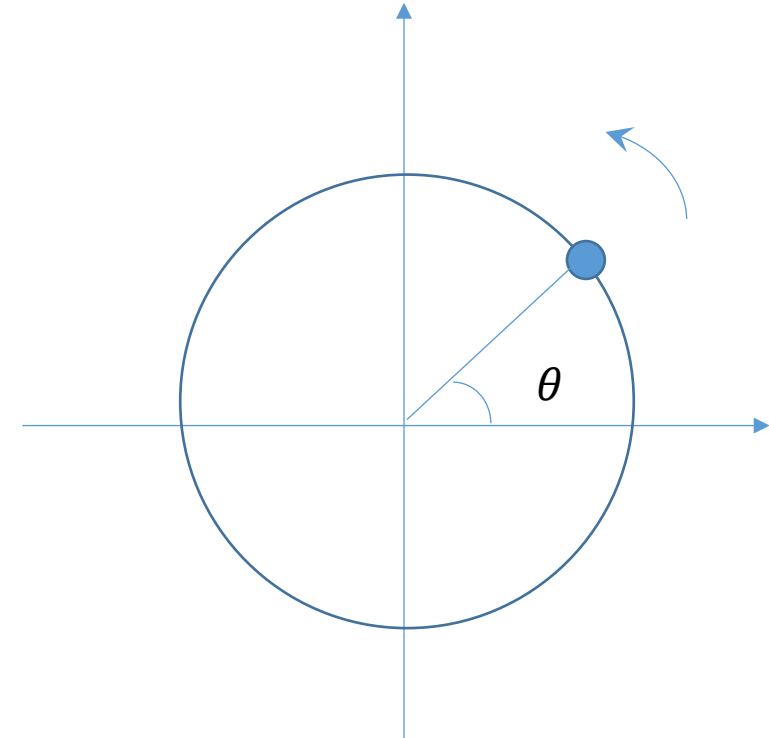
Rotation about a fixed axis



https://en.wikipedia.org/wiki/Rotation_around_a_fixed_axis

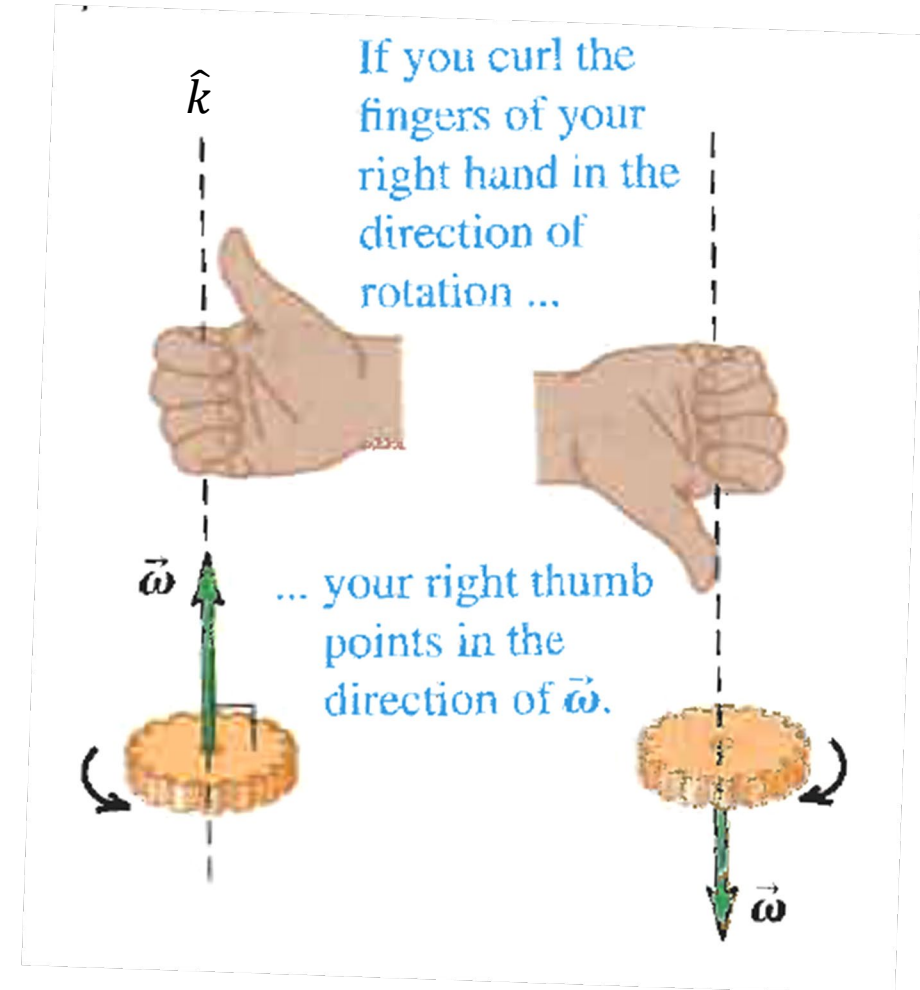
Description of Rotational Motion

- In translational motion, we describe the motion of a moving object in terms of the concepts of position, displacement, velocity and acceleration.
- How do we describe rotational motion about a fixed axis?
- In terms of angle variables, angular position, angular displacement, angular velocity and angular acceleration!



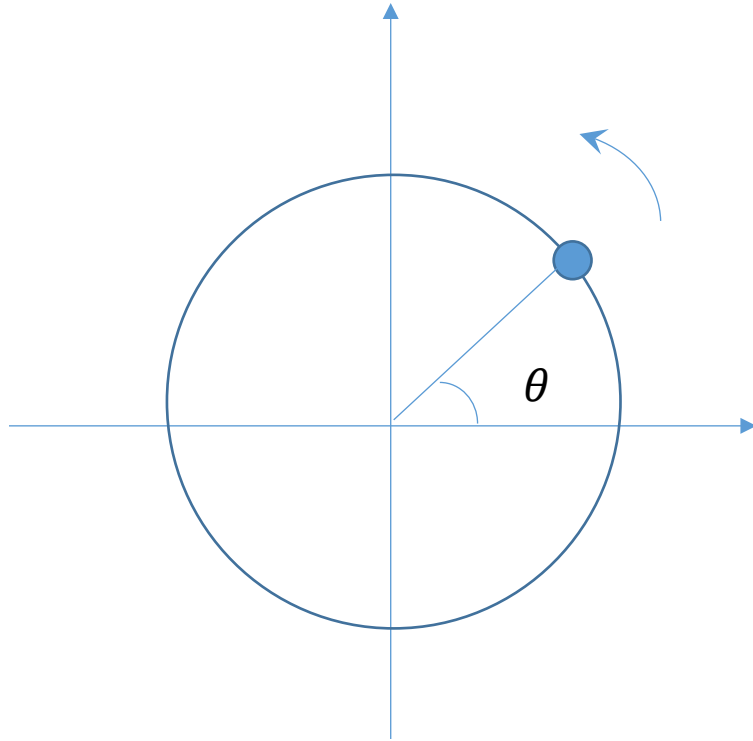
Description of Rotational Motion

- The direction of rotational motion is defined via vector direction of rotation (described by $\hat{k}$) and the actual rotation is determined according to the right hand rule.



H. D. Young and R. A. Freedman, University Physics (14th edition), p300

Description of Rotational Motion



- Angular position, θ (rad)
- Angular velocity (rad/s):

$$\vec{\omega} = \omega_z \hat{k} \quad \text{with} \quad \omega_z = \frac{d\theta}{dt}$$

- Angular acceleration (rad/s²):

$$\vec{\alpha} = \alpha_z \hat{k} \quad \text{with} \quad \alpha_z = \frac{d\omega_z}{dt} = \frac{d^2\theta}{dt^2}$$

H. D. Young and R. A. Freedman, University Physics (14th edition), p298-302

Fixed-Axis Rotation for Point Particle

- Kinematics**

Angular position: θ

Angular velocity: $\vec{\omega} = \omega_z \hat{k}$

$$\omega_z = \frac{d\theta}{dt} \quad \begin{array}{l} \omega_z > 0, \text{direction} + \hat{k} \\ \omega_z < 0, \text{direction} - \hat{k} \end{array}$$

Angular acceleration: $\vec{\alpha} = \alpha_z \hat{k}$

$$\alpha_z = \frac{d\omega_z}{dt} \quad \begin{array}{l} \alpha_z > 0, \text{direction} + \hat{k} \\ \alpha_z < 0, \text{direction} - \hat{k} \end{array}$$

Angular acceleration: $\vec{\alpha} = \alpha_z \hat{k}$

Angular velocity: $\vec{\omega} = \omega_z \hat{k}$

$$\omega_z(t_1) = \omega_{z,0} + \int_{t_0}^{t_1} \alpha_z(t) dt$$

$$\begin{array}{l} \omega_z > 0, \text{direction} + \hat{k} \\ \omega_z < 0, \text{direction} - \hat{k} \end{array}$$

Angular position: θ

$$\theta(t_1) = \theta_0 + \int_{t_0}^{t_1} \omega_z(t) dt$$

Fixed-Axis Rotation for Point Particle: constant α_z

- Kinematics**

Angular position : θ

Angular velocity: $\vec{\omega} = \omega_z \hat{k}$

$$\omega_z = \frac{d\theta}{dt} \quad \begin{array}{l} \omega_z > 0, \text{ direction } + \hat{k} \\ \omega_z < 0, \text{ direction } - \hat{k} \end{array}$$

Angular acceleration: $\vec{\alpha} = \alpha_z \hat{k}$

$$\alpha_z = \frac{d\omega_z}{dt} \quad \begin{array}{l} \alpha_z > 0, \text{ direction } + \hat{k} \\ \alpha_z < 0, \text{ direction } - \hat{k} \end{array}$$

Angular acceleration: $\vec{\alpha} = \alpha_z \hat{k}$

Angular velocity: $\vec{\omega} = \omega_z \hat{k}$

$$\omega_z = \omega_{z0} + \alpha_z t \quad \begin{array}{l} \omega_z > 0, \text{ direction } + \hat{k} \\ \omega_z < 0, \text{ direction } - \hat{k} \end{array}$$

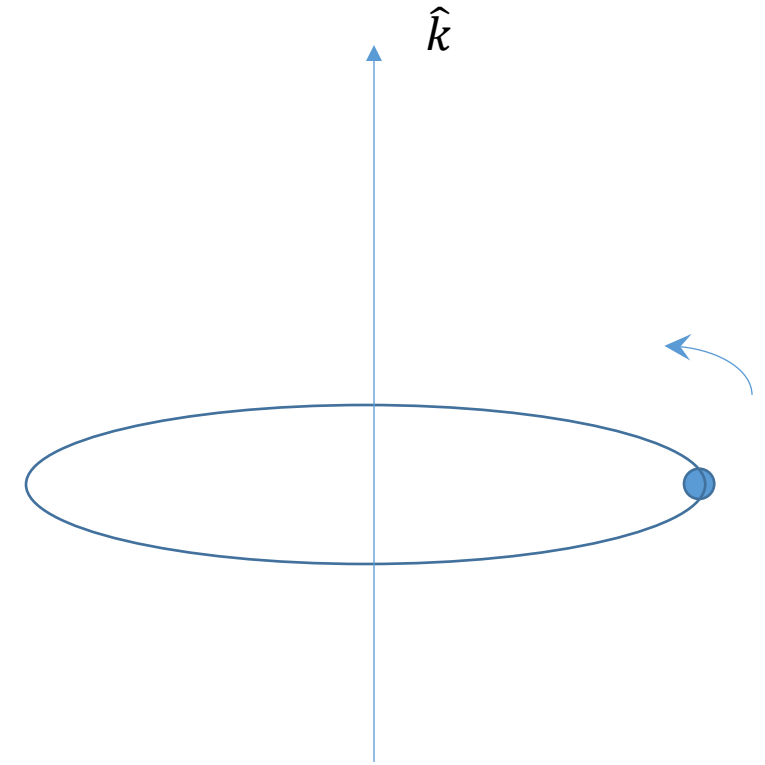
Angular position: θ

$$\theta = \theta_0 + \omega_{z0}t + \frac{1}{2}\alpha_z t^2$$

Concept Question

A point particle is rotating as shown below. If the particle experiences an angular acceleration of $\vec{\alpha} = -\alpha_z \hat{k}$, what will happen to rotation of the particle at later time?

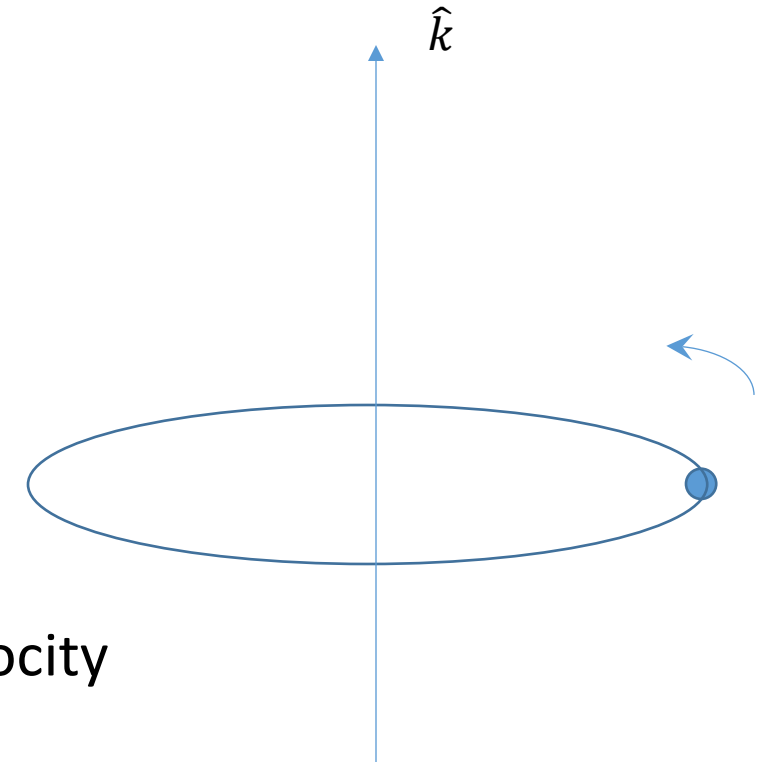
1. The wheel will speed up
2. The wheel will slow down
3. The wheel will move at the same speed
4. No enough information



Concept Question - Solution

A point particle is rotating as shown below. If the particle experiences an angular acceleration of $\vec{\alpha} = -\alpha_z \hat{k}$, what will happen to rotation of the particle at later time?

1. The wheel will speed up
2. The wheel will slow down
3. The wheel will move at the same speed
4. No enough information



Answer (2). The particle will slow down since the angular velocity and the angular acceleration are in different directions.

Concept Question

A flywheel is rotating with $\theta(t) = 3t - t^2$ (rads), selecting “into the page” as the positive vector direction of rotation. Find the direction of angular velocity at $t = 1$ s.

1. Into the page
2. Out of the page
3. Up
4. Down
5. Zero

Concept Question - Solution

A flywheel is rotating with $\theta(t) = 3t - t^2$ (rads), selecting “into the page” as the positive vector direction of rotation. Find the direction of angular velocity at $t = 1$ s.

1. Into the page
2. Out of the page
3. Up
4. Down
5. Zero

Answer (1).

Concept Question

A flywheel is rotating with $\theta(t) = 3t - t^2$ (rads), selecting “into the page” as the positive vector direction of rotation. Find the direction of angular acceleration at $t = 1$ s.

1. Into the page
2. Out of the page
3. Up
4. Down
5. Zero

Concept Question - Solution

A flywheel is rotating with $\theta(t) = 3t - t^2$ (rads), selecting “into the page” as the positive vector direction of rotation. Find the direction of angular acceleration at $t = 1$ s.

1. Into the page
2. Out of the page
3. Up
4. Down
5. Zero

Answer (2).

Question

A flywheel is rotating about z axis (and the positive vector direction of rotation in $\hat{k}$), with an angular velocity of 600 rad/s . Then an external brake is exerted on the flywheel at a rate of 25 rad/s^2 .

- 1) Find the time required for the flywheel to stop.
- 2) Find the angular displacement of the flywheel.
- 3) Find the number of rotations completed by the flywheel.

Question - Solution

A flywheel is rotating about z axis (and the positive vector direction of rotation in $\hat{k}$), with an angular velocity of 600 rad/s. Then an external brake is exerted on the flywheel at a rate of 25 rad/s².

- 1) Find the time required for the flywheel to stop.
- 2) Find the angular displacement of the flywheel.
- 3) Find the number of rotations completed by the flywheel.

$$\omega_z = \omega_{z0} + \alpha_z t \Rightarrow 0 = 600 - 25 t \Rightarrow t = 24 \text{ s}$$

$$\theta = \theta_0 + \omega_{z0} t + \frac{1}{2} \alpha_z t^2 \Rightarrow \theta = 600 t - \frac{1}{2} 25 t^2 = 7200 \text{ rad}$$

$$\text{Number of rotations: } \frac{7200}{2\pi} = 1145.92$$

Question

A flywheel is rotating about z axis, with its angular position described by $\theta(t) = 3t + 2 \cos t$, and the positive vector direction of rotation in $\hat{k}$.

- 1) What is the angular velocity of the flywheel? Does the angular velocity change direction?
- 2) What is the angular acceleration of the flywheel? Does the angular acceleration change direction?

Question - Solution

A flywheel is rotating about z axis, with its angular position described by $\theta(t) = 3t + 2\cos t$, and the positive vector direction of rotation in $\hat{k}$.

- 1) What is the angular velocity of the flywheel? Does the angular velocity change direction?
- 2) What is the angular acceleration of the flywheel? Does the angular acceleration change direction?

$$\omega_z = \frac{d\theta}{dt} = 3 - 2\sin t \quad \longrightarrow \quad \vec{\omega} = (3 - 2\sin t) \hat{k} \quad \text{No change in direction}$$

$$\alpha_z = \frac{d\omega_z}{dt} = -2\cos t \quad \longrightarrow \quad \vec{\alpha} = -2\cos t \hat{k} \quad \text{Direction is changing with } t$$

Fixed-Axis Rotation

- Torque about an axis of rotation produces angular acceleration about the axis

$$\vec{\tau} = I_s \vec{\alpha}$$

I_s is the moment of inertia about the axis of rotation

$\vec{\alpha}$ is the angular acceleration about the axis of rotation

- Analogous to Newton's Second Law for Linear Motion

$$\vec{F} = m\vec{a}$$

http://physics.bu.edu/~redner/211-sp06/class13/class14_newton2.html

Moment of inertia

- Moment of inertia describes the resistance of an object to any change in its state of rotation.

$$\vec{\tau} = I_s \vec{\alpha}$$

- Analogous to mass for translational motion

$$\vec{F} = m \vec{a}$$

<https://www.physicsclassroom.com/class/newtlaws/Lesson-1/Inertia-and-Mass>

<https://opentextbc.ca/universityphysicsv1openstax/chapter/10-4-moment-of-inertia-and-rotational-kinetic-energy/>

Moment of inertia

For a point object (such as a ball of mass m) rotating about an axis of rotation s , the moment of inertia of the object about the axis is

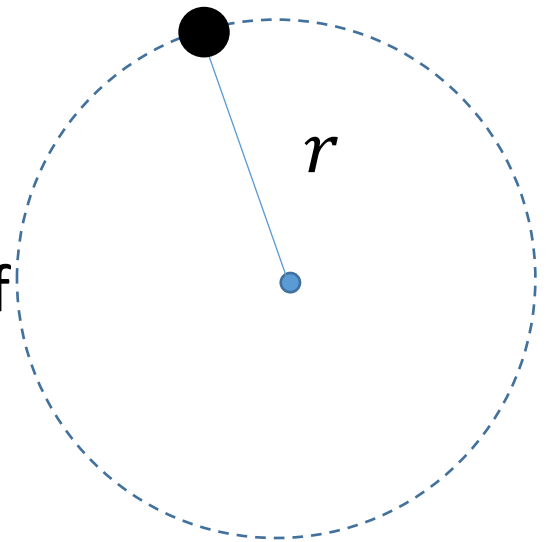
$$I_s = mr^2 \quad (\text{kg} \cdot \text{m}^2)$$

where r is the distance of the object from the axis of rotation.

For a system composed of n point objects rotating about an axis of rotation s , the moment of inertia of the system about the axis is

$$I_s = \sum_{i=1}^n m_i r_i^2 \quad (\text{kg} \cdot \text{m}^2)$$

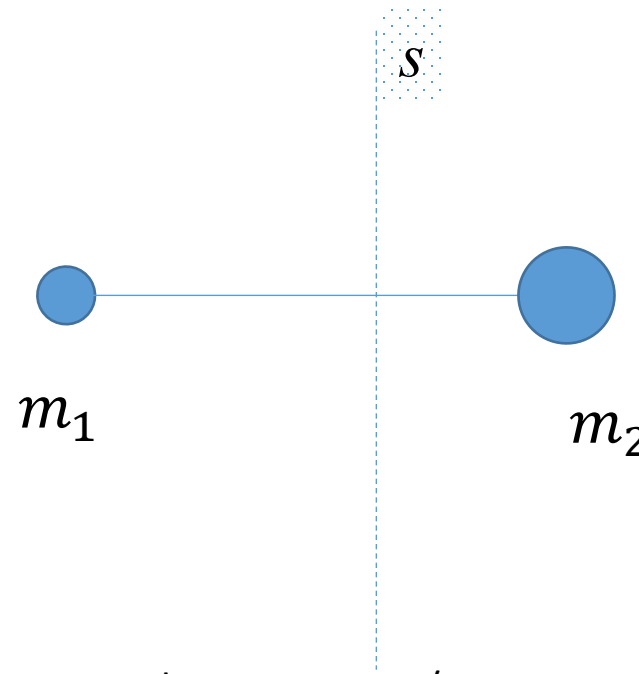
where r_i is the distance of the object with m_i from the axis of rotation.



<https://www.khanacademy.org/science/physics/torque-angular-momentum/torque-tutorial/a/rotational-inertia>

Question

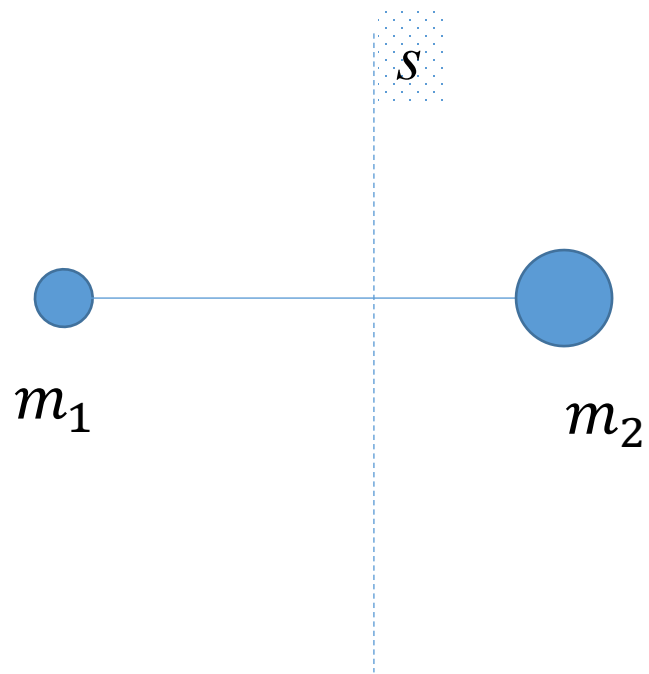
There are two balls of mass $m_1 = 0.5 \text{ kg}$ and $m_2 = 2 \text{ kg}$. The two balls are each attached to one end of a massless rod, as shown. m_1 is 0.2 m away from the axis s and m_2 is 0.1 m away from the axis. When the whole system is rotating about the axis s , find the moment of inertia of the system.



<https://www.khanacademy.org/science/physics/torque-angular-momentum/torque-tutorial/a/rotational-inertia>

Question - Solution

There are two balls of mass $m_1 = 0.5 \text{ kg}$ and $m_2 = 2 \text{ kg}$. The two balls are each attached to one end of a massless rod, as shown. m_1 is 0.2 m away from the axis and m_2 is 0.1 m away from the axis. When the whole system is rotating about axis s , find the moment of inertia of the system.



$$\begin{aligned} I_s &= \sum_{i=1}^n m_i r_i^2 \\ &= m_1 r_1^2 + m_2 r_2^2 = 0.04 \text{ kg} \cdot \text{m}^2 \end{aligned}$$

Moment of inertia

For objects that cannot be approximated by point objects,

$$I_s = \int_{\text{object}} r^2 dm$$

<https://opentextbc.ca/universityphysicsv1openstax/chapter/10-5-calculating-moments-of-inertia/>

https://en.wikipedia.org/wiki/List_of_moments_of_inertia

<https://www.khanacademy.org/science/physics/torque-angular-momentum/torque-tutorial/a/rotational-inertia>

Question

- 1) A flywheel with a moment of inertia of $15 \text{ kg} \cdot \text{m}^2$, is rotating from rest to an angular velocity of 40 rad/s in 20 seconds at a constant rate. Find the net torque exerted on the flywheel to cause the rotation.
- 2) By changing the forces acting on the flywheel, the flywheel then comes to stop in 10 seconds at a constant rate. Find the net torque experienced by the flywheel.

Question - Solution

- 1) A flywheel with a moment of inertia of $15 \text{ kg} \cdot \text{m}^2$, is rotating from rest to an angular velocity of 40 rad/s in 20 seconds at a constant rate. Find the net torque exerted on the flywheel to cause the rotation.
- 2) By changing the forces acting on the flywheel, the flywheel then comes to stop in 10 seconds at a constant rate. Find the net torque experienced by the flywheel.

$$\omega_z = \omega_{z0} + \alpha_z t \Rightarrow 40 = 0 + \alpha_z \times 20 \Rightarrow \alpha_z = 2 \text{ rad/s}^2$$

$$\tau = I_s \alpha = 15 \times 2 = 30 \text{ N} \cdot \text{m}$$

$$\omega'_z = \omega'_{z0} + \alpha'_z t' \Rightarrow 0 = 40 - \alpha'_z \times 10 \Rightarrow \alpha'_z = 4 \text{ rad/s}^2$$

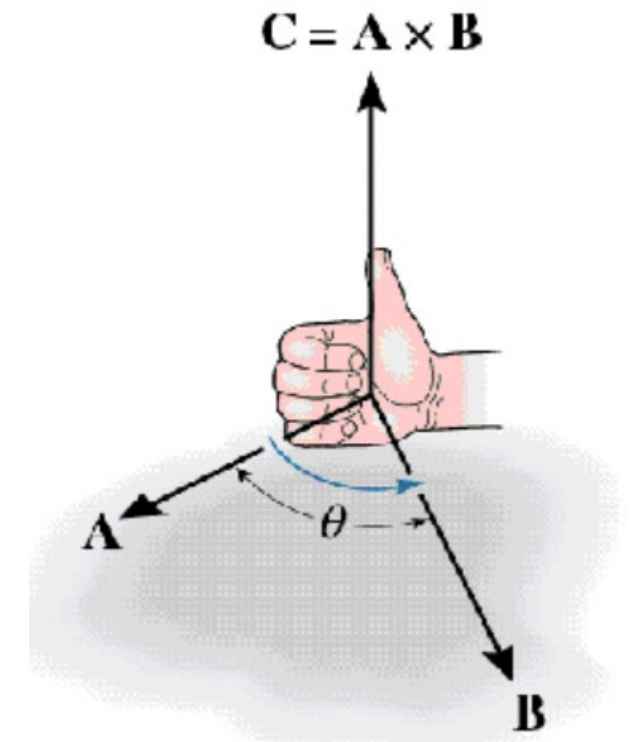
$$\tau' = I_s \alpha' = 15 \times 4 = 60 \text{ N} \cdot \text{m}$$

Vector (Cross) Product $\vec{A} \times \vec{B}$

- **Magnitude**:

$$|\vec{A} \times \vec{B}| = AB \sin \theta \quad \text{where } \theta \text{ is the angle between the two vectors.}$$

- **Direction**: determined by the Right-Hand-Rule



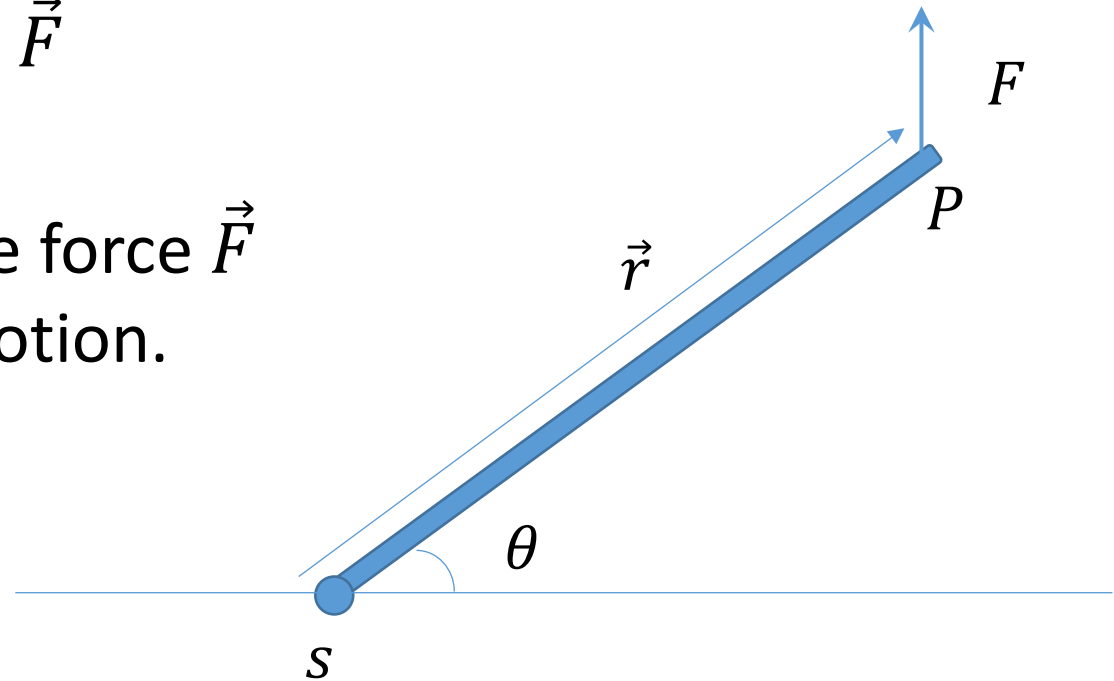
<https://web.aeromech.usyd.edu.au/statics/doc/math3.htm>

Torque: Magnitude and Direction

- Given a force $\vec{F}$ exerted on an object at a point P whose position is described by $\vec{r}$ with respect to the pivot of rotation s , we then define the torque due to the force $\vec{F}$ about the pivot s as,

$$\vec{\tau}_s = \vec{r} \times \vec{F}$$

- Torque measures the ability of the force $\vec{F}$ to cause changes in rotational motion.
- SI unit: $\text{N} \cdot \text{m}$

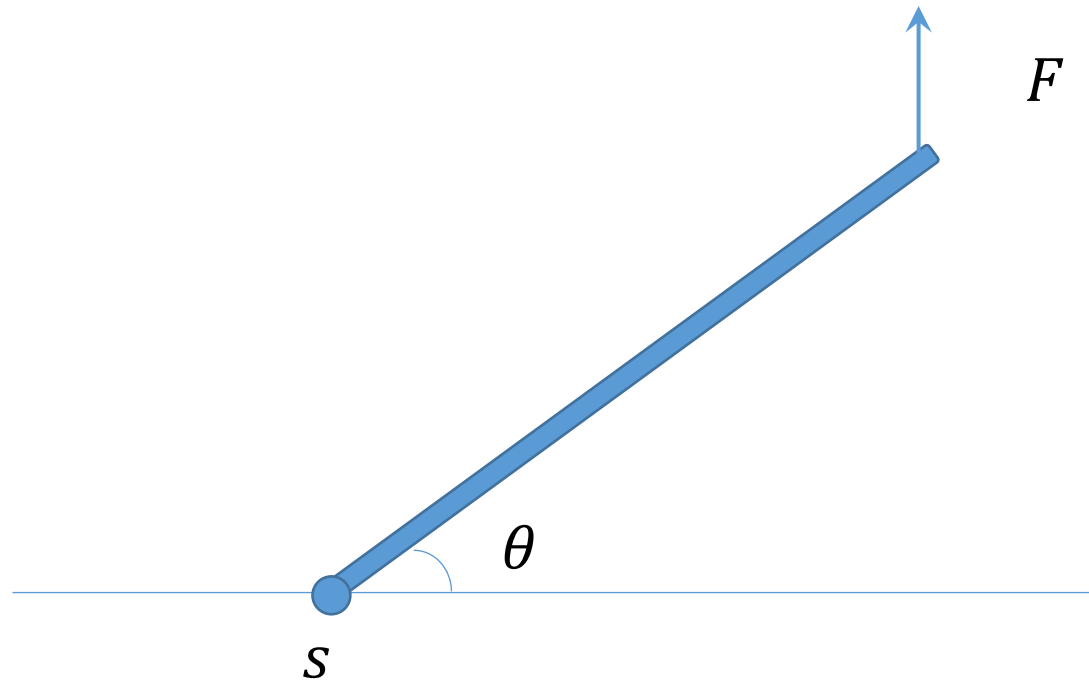


<https://en.wikipedia.org/wiki/Torque>

Concept Question

A force of magnitude F is exerted on one end of a rod of length l , as shown. Find the magnitude of the torque applied by F about the point s .

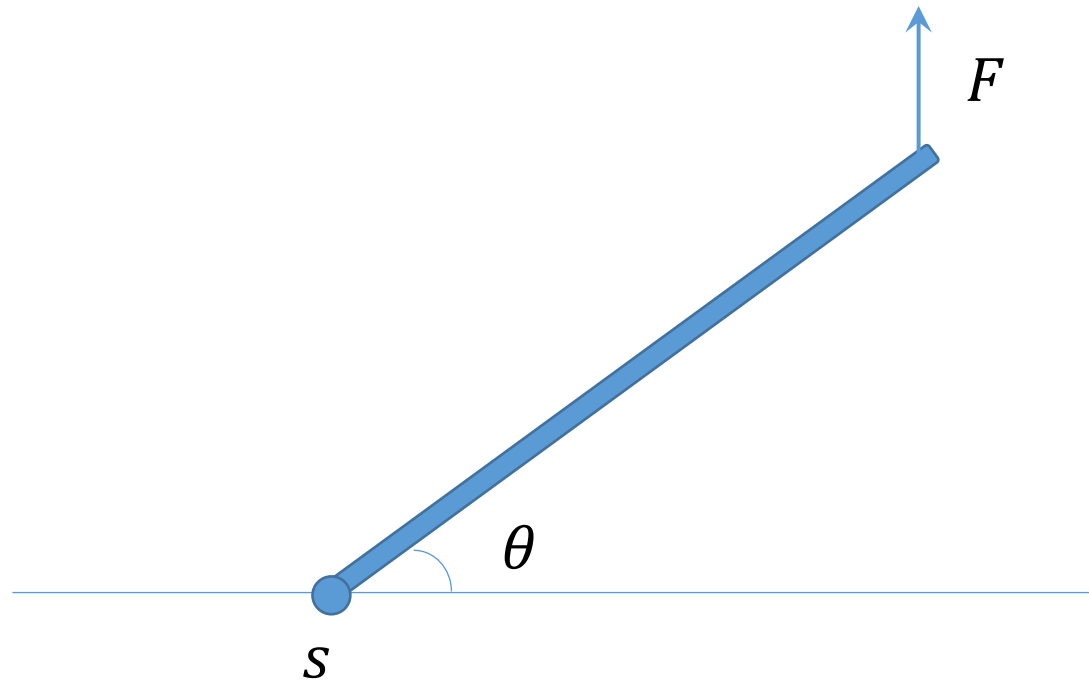
1. $lF \sin \theta$
2. $lF \cos \theta$
3. lF
4. 0



Concept Question - Solution

A force of magnitude F is exerted on one end of a rod of length l , as shown. Find the magnitude of the torque applied by F about the point s .

1. $lF \sin \theta$
2. $lF \cos \theta$
3. lF
4. 0

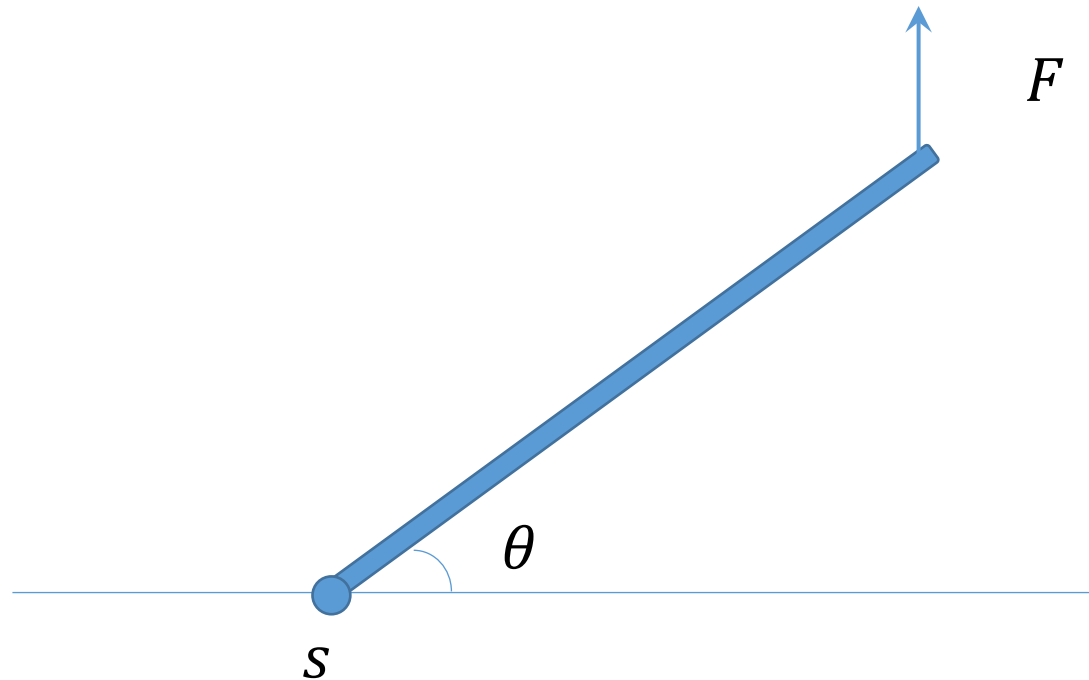


Answer: (2)

Concept Question

A force of magnitude F is exerted on one end of a rod of length l , as shown. Find the direction of the torque applied by F about the point s .

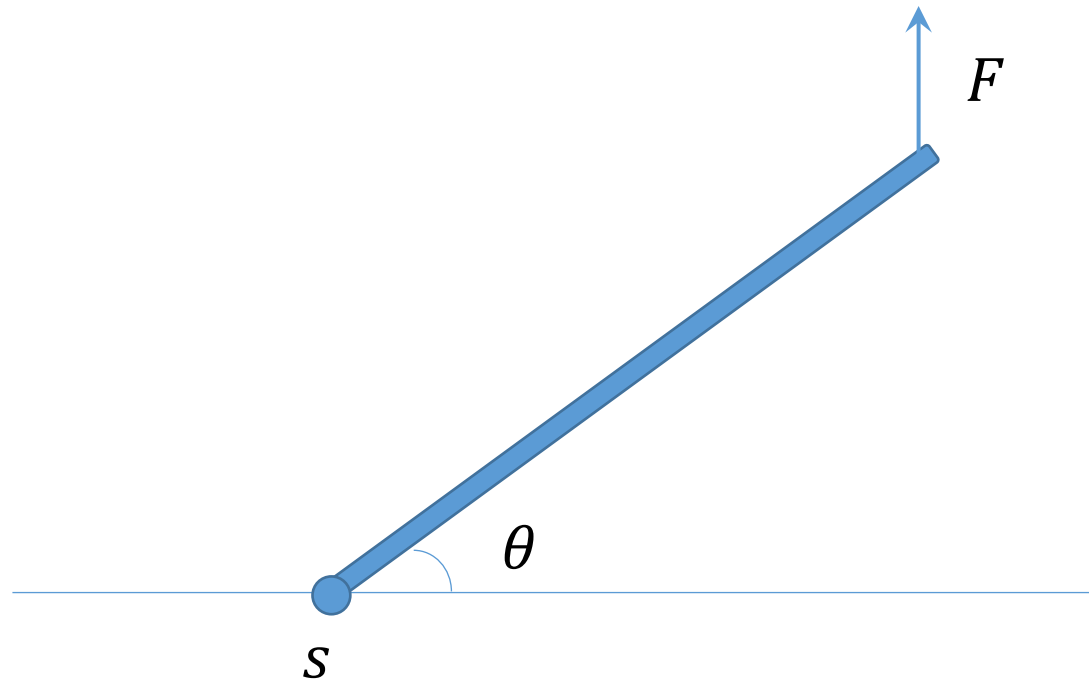
1. Out of the plane
2. Into the plane
3. Up
4. Down



Concept Question - Solution

A force of magnitude F is exerted on one end of a rod of length l , as shown. Find the direction of the torque applied by F about the point s .

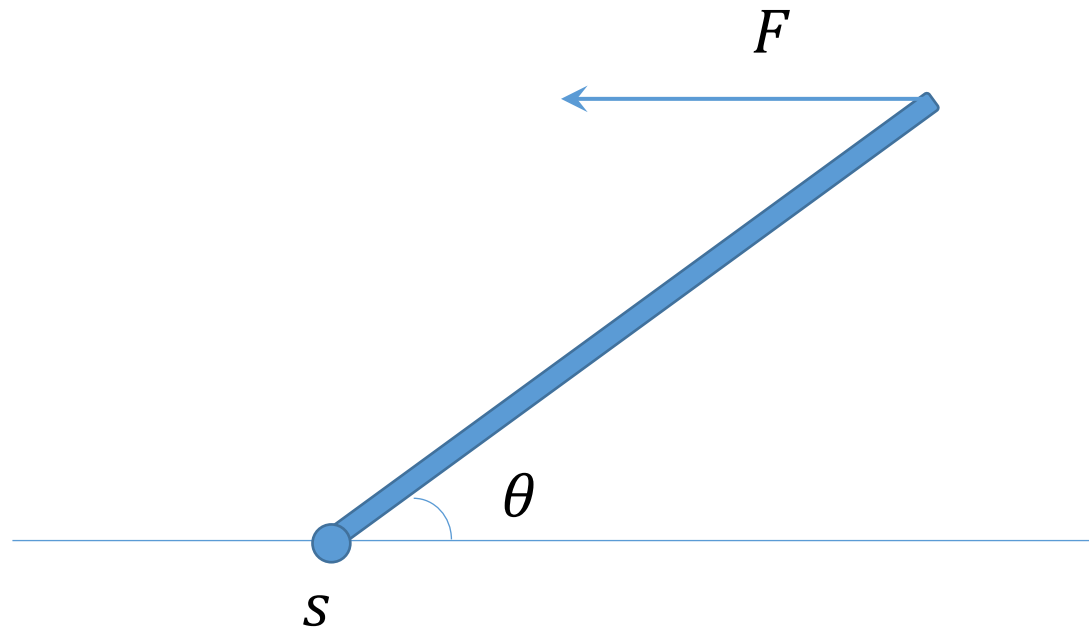
1. Out of the plane
2. Into the plane
3. Up
4. Down



Answer: (1)

Example

A force of magnitude F is exerted on one end of a rod of length l and mass m , as shown. Find the net torque applied on the rod about the point s .



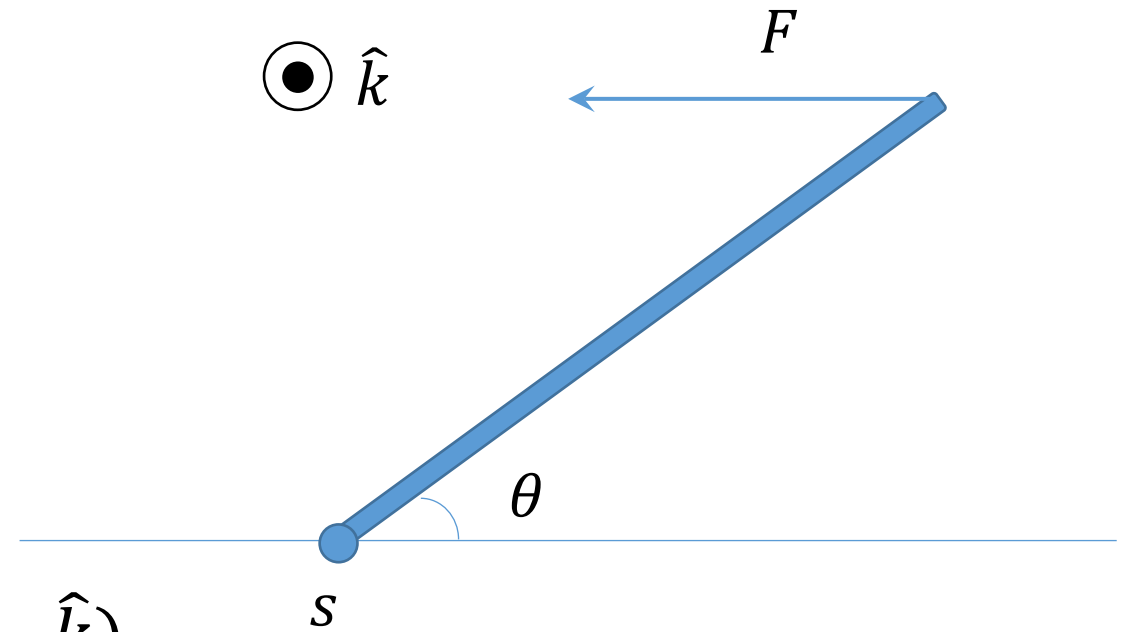
Example - Solution

A force of magnitude F is exerted on one end of a rod of length l and mass m , as shown. Find the net torque applied on the rod about the point s .

$$\tau_F = Fl \sin(180^\circ - \theta) = Fl \sin \theta$$

$$\tau_G = \frac{mgl}{2} \sin(\theta + 90^\circ) = \frac{mgl}{2} \cos \theta$$

$$\vec{\tau} = \vec{\tau}_F + \vec{\tau}_G = Fl \sin \theta \hat{k} + \frac{mgl}{2} \cos \theta (-\hat{k})$$



Off-class practice question(s)

Concept Question

A flywheel is rotating with $\theta(t) = 3t - t^2$ (rads), selecting “into the page” as the positive vector direction of rotation. Find the direction of angular velocity at $t = 2\text{s}$.

1. Into the page
2. Out of the page
3. Up
4. Down
5. Zero

Concept Question - Solution

A flywheel is rotating with $\theta(t) = 3t - t^2$ (rads), selecting “into the page” as the positive vector direction of rotation. Find the direction of angular velocity at $t = 2\text{s}$.

1. Into the page
2. Out of the page
3. Up
4. Down
5. Zero

Answer (2).

Concept Question

A flywheel is rotating with $\theta(t) = 3t - t^2$ (rads), selecting “into the page” as the positive vector direction of rotation. Find the direction of angular acceleration at $t = 2\text{s}$.

1. Into the page
2. Out of the page
3. Up
4. Down
5. Zero

Concept Question - Solution

A flywheel is rotating with $\theta(t) = 3t - t^2$ (rads), selecting “into the page” as the positive vector direction of rotation. Find the direction of angular acceleration at $t = 2\text{s}$.

1. Into the page
2. Out of the page
3. Up
4. Down
5. Zero

Answer (2).